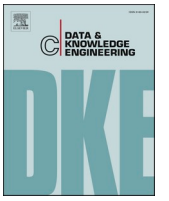
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# A survey of processing methods for different types of concept drift

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ABSTRACT

For large, dynamic and continuously changing data streams, the characteristics and categories of data samples change with the increase of time, which is called concept drift. Concept drift has an impact on the stability and performance of the model, if ignored, the prediction will be inaccurate. More and more attention has been paid to the study of multi-data streams. Group concept drift is more complicated than single data stream concept drift. This paper studies the processing methods of group concept drift, and classifies the processing methods from three directions of single type, multiple type and group type for the first time. Within the scope of methods designed to address a single type of concept drift, this paper provides a systematic summary based on the classification of drift types, distinguishing between approaches that target real concept drift and those that address virtual concept drift. Solving real concept drift includes abrupt, gradual, recurrent and local concept drift processing methods. In this paper, the methods of dealing with the concept drift of multiple type are introduced from the aspects of solving the drift of gradual and abrupt, abrupt, gradual and incremental, abrupt, gradual and recurrent. This paper analyzes and summarizes the methods to solve the group type concept drift problem. The methods considered in this paper are compared and summarized from the aspects of processing technology, applicable drift type, comparison methods, advantages and disadvantages.

## 1. Introduction

In the big data environment, more and more data need to be processed. In today's digital age, data is one of the key factors driving science, business and society [1]. The development of data analysis and machine learning technology has provided people with the ability to extract knowledge and make predictions from massive amounts of data. However, with the passage of time and the change of environment, the concepts contained in a single data stream also continue to evolve, a phenomenon called Concept Drift [2]. In multi-data streams, the simultaneous occurrence of correlated concept drift is called Group Concept Drift [3]. This change may be due to external factors, such as changes in market trends or changes in user behavior, or it may be caused by biases or noise during data collection. Concept drift can occur in a variety of fields and applications, including fraud detection [4], financial market forecasting [5], medical diagnostics [6], intelligent transportation systems [7], etc. In practical applications, the occurrence of concept drift is usually detected by monitoring the performance change of the model over a continuous period of time. If the performance of the model

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varies significantly over different time periods, there may be concept drift.

The methods to solve concept drift can be divided into active detection method and passive adaptation method. The classic Drift Detection Method (DDM) [8] based on error rate shows that the class distribution has changed and the actual decision model is not suitable because the error of the algorithm increases significantly. Therefore, when the error rate is greater than a certain threshold, it means that drift occurs. Early Drift Detection Method (EDDM) [9] deems that conceptual drift occurs when the time interval between two classification errors is significantly reduced. This method improves the detection of gradual drift on the basis of DDM, but is not good at data sets containing noise. The Statistical Test of Equal Proportions (STEPD) [10], based on statistical tests, considers recent accuracy and overall accuracy as two equally important criteria. If the accuracy is significantly reduced recently, drift occurs. This method is fast and accurate, but due to its overly sensitive monitoring of changes, there are also false detections. Adaptive Windowing (ADWIN) [11] determines whether drift occurs by monitoring the error rate of the current model. This method reduces false positives and missed positives, but has low temporal and spatial efficiency. In the detection process, the multi-sliding window-based (CDT_MSW) [12] method detects the drift position by sliding a single fixed-size window, determines the drift category by detecting the drift length with the companion window, and uses multiple sliding Windows to track and obtain key information to identify the drift subcategory. The group drift detection method (GDDM) [3] for multi-data streams is based on error rate like DDM, and three window types of sliding window, initial window and detection window are used simultaneously to consider the online error rate of all data streams. The above is a concept Drift detection method in supervised environment. In unsupervised environment, there is a Label Dependency Drift Detector (LD3) [13] which utilizes the dynamic time dependence between labels, uses the label influence ranking method, and uses the data fusion algorithm. The resulting rankings are used to detect concept drift. Statistical Drift Detection Method (SDDM) based on data distribution [14] confirms whether drift occurs by detecting changes in data distribution. This method has fewer false positives and false negatives, but only considers part of the feature distribution. Passive adaptation methods mainly include effective allocation examples among classifiers based on Adaptable diversity-based Online Boosting (ADOB) [15], which controls the Diversity execution distribution by the accuracy of each classifier. Diverse instance-weighting Ensemble (DiwE) [16] is tested using a diversity measurement based on whether set members agree with the probability of regional distribution changes, and the estimation of regional distribution changes is regarded as instance weights. Constructing different sets of regions through different schemes will lead to different drift estimation results, resulting in diversity, while choosing the classifier with the most disagreement to maximize diversity.

In recent years, in a review of concept drift processing methods, Gama et al. [17] introduced the process of adaptive learning in a supervised environment, comprehensively covering algorithms based on memory, change detection, learning, and loss estimation. Iwashita et al. [18] considered different types of drift and corresponding handling methods, taking into account factors such as the choice of learners in classification tasks and the use of drift detection mechanisms, and summarized categories such as weight-based, instance selection, window-based, and classifier integration methods. Sato et al. [19] analyzed active drift detection methods

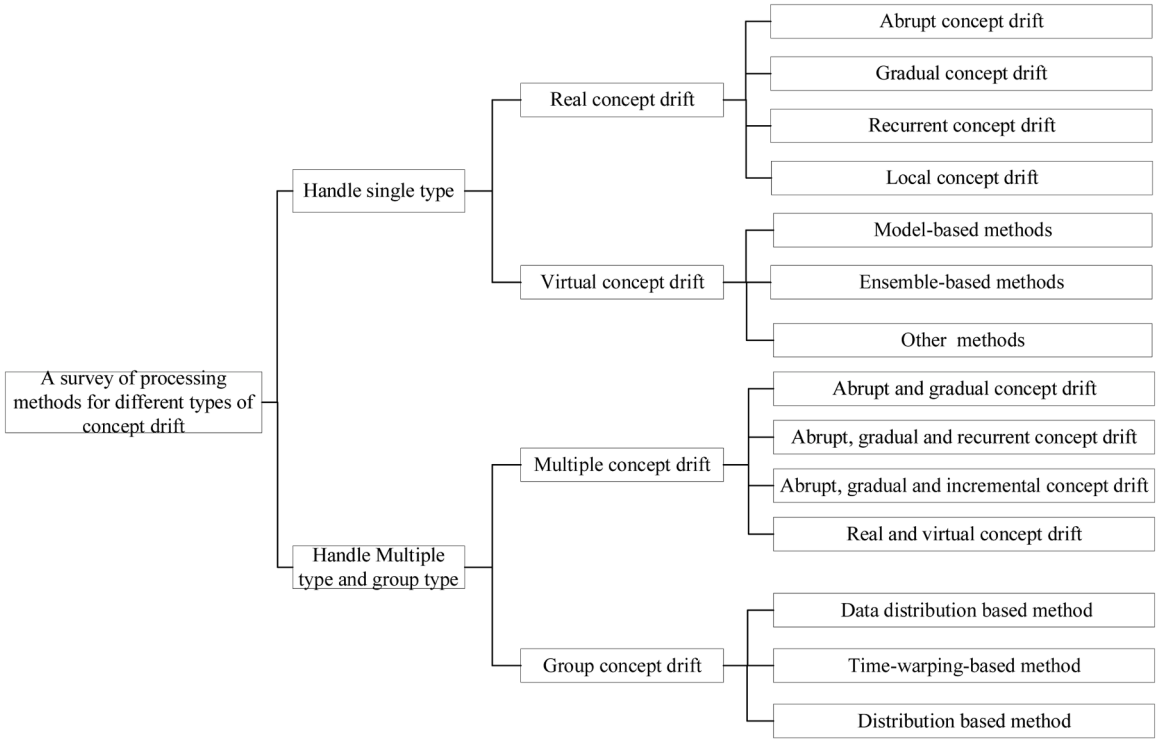



**Fig. 1.** Overall frame diagram.

according to different drift types. Agrahari et al. [20] discussed concept drift detectors for semi-supervised and unsupervised data, highlighting their key features, advantages, disadvantages, and outlining future research challenges. Han et al. [21] explained concept drift processing methods from the two directions of active processing and passive processing. The active processing method was explained from the processing of specific types and multiple types, and the passive processing method was explained from the single classifier and ensemble learning. Trat et al. [22] focused on the integration of concept drift detection, providing an overview and synthesis of related approaches. In summary, previous surveys have mainly classified concept drift processing methods into active versus passive approaches or into single versus complex drift scenarios. Distinct from these works, this paper introduces group type drift as an additional category, providing a systematic review from the three perspectives of single type, multiple type, and group type drift. This classification highlights an underexplored dimension of drift research, thereby complementing prior reviews and clarifying our unique contribution. The main contributions are as follows:

1) Classify the concept drift processing methods from a new perspective, and analyze the concept drift processing methods from single type concept drift, multiple type concept drift and group type concept drift.
2) The processing method of a single type is analyzed from the two aspects of solving real drift and virtual drift. The real concept drift is divided into abrupt drift, gradual drift, recurrent drift and other real types of drift; The processing of complex types is analyzed from abrupt drift and gradual drift, abrupt, gradual and incremental, abrupt, gradual and recurrent, and other complex types.
3) Summarize the existing concept drift processing methods, analyze the existing problems in the current research situation of concept drift processing methods, give the future research direction, and provide new ideas.

The rest of this paper is organized as follows: Section 2 introduces the related concepts and principles used in this paper. The third section introduces the concept drift processing methods for dealing with a single type and makes a comparative analysis. The fourth section introduces the concept drift processing method for complex type and group type, and gives a systematic overview. The fifth section analyzes the next research direction. The sixth section summarizes. The overall framework of this paper is shown in Fig. 1.

## 2. Related work

### 2.1. Concept drift

In a stationary data stream, the data distribution is stable, so a valid model is still valid for new data. In a non-stationary data stream, the relationship between the characteristics of a data sample and the category to which it belongs may change over time, resulting in a degradation of the performance of a previously valid model on new data called concept drift, as shown in Definition 1 [23]. With the increase of time, the joint concept distribution of the instance feature vector and the target class label in the data stream is no longer equal, that is, the data distribution of the data sample changes, which means that concept drift occurs.

**Definition 1.** (Concept drift [23]). Suppose that a continuous instance in a data stream appears in the form of $\left(x_j, y_j\right)$, where $x_j$ represents the feature vector and $y_j$ is a collection of $m$ class labels, that is $y \in \left\{y_1, y_2, \ldots, y_m\right\}$, and $j = 1, 2, 3, \ldots$. $P_j$ represents the joint probability distribution between eigenvector $x_j$ and the target class label $y_j$ at time $t$. Conceptual drift at the moment of arrival is shown in formula Eq. (1):

$$\exists x_j : P_{j_0}\left(x_j, y_j\right) \neq P_{j_1}\left(x_j, y_j\right) \tag{1}$$

There are many classification methods of concept drift. This paper classifies it from the source, velocity and spatial distribution of drift. Firstly, according to the influence of class conditional probability distribution on class posterior probability distribution, Concept Drift can be divided into Real Concept Drift [24] and Virtual Concept Drift [25]. True concept drift [26] means that the class posterior probability distribution changes regardless of whether the prior probability changes, thus affecting the decision boundary of the classifier and reducing the performance of the learner. This type of drift is usually caused by a change in the external environment or a change in the task itself. Virtual concept drift [27] refers to when the input distribution or class distribution changes, but it is not affected, and therefore the decision boundary is not affected. Unlike real concept drift, virtual concept drift is not due to the change of the concept itself generated by the data, but by manipulating the data or introducing noise to simulate the phenomenon of drift. Real concept drift and virtual concept drift are shown in Fig. 2.

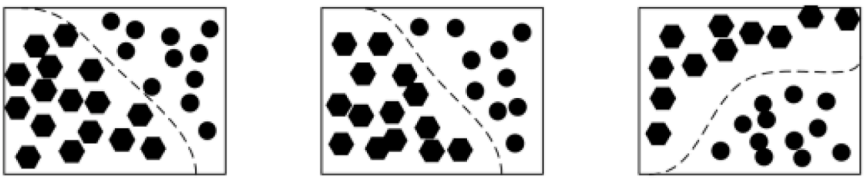



**Fig. 2.** Real and virtual concept drift.

According to the speed and duration of concept change, concept drift can be divided into gradual drift, abrupt drift, incremental drift and recurrent drift [28]. Abrupt concept drift [29] refers to a situation in which the concept changes dramatically. In this case, the feature distribution or category label of the data may abruptly change significantly at some point in time, rather than evolve gradually. Gradual conceptual drift [30] means that the feature distribution or category label of data may change gradually and continuously, rather than abruptly. This type of drift lasts for a long time, and the speed of change is relatively slow. Recurrent concept drift refers to the situation in which concept drift occurs periodically. In this case, the feature distribution or category label of the data may change according to a certain periodic pattern, rather than continuous change. Incremental concept drift [6] means that the birth of concept drift is a process of gradual accumulation and gradual increase. Incremental concept drift is a gradual drift phenomenon, and its influence accumulates and increases gradually. The schematic diagram of the classification of real concept drift according to drift speed and duration is shown in Fig. 3.

Global concept drift and local concept drift are classified according to the range or influence of drift [31]. They describe the propagation mode and influence degree of concept drift in data distribution [32]. Global concept drift refers to the situation in which concept drift affects the whole data set or the overall data distribution. Global concept drift may be caused by abrupt changes in the external environment or major changes in the data generation process. Local concept drift refers to the situation in which concept drift affects only part of the sample or local area of the data set [33]. Global concept drift and local concept drift are shown in Fig. 4.

Usually in a single data stream, there is only one type of concept drift, but some single data streams may contain multiple types of concept drift types, that is, composite types. The above descriptions are all for a single data stream, but real-world applications usually involve multiple mixed data streams, in which a large number of marked and unmarked data streams arrive at the same time, and the concept drift spans multi-data streams. and in the case that the drift of each data stream cannot be detected in time, a slight underlying distribution drift is called group type concept drift, as shown in Definition 2 [3]. If, over time, the instances in the data stream no longer follow a particular distribution $E_{(0,j)}(G_i)$, that is, the distribution of the instances $G_{(j,j+1)}$ has changed, it represents a group concept drift.

**Definition 2.** (Group concept drift [3]): suppose there are $M$ data stream, and in time $[0, j]$, a set of instances is expressed as $G_{0,j} = G_0, \ldots G_i, \ldots G_j$, where $G_i = (g_1, \ldots g_i, \ldots g_M)$, $g_i = (X_i, Y_i)$ is the observation obtained from the article $i$ data stream, $X_i$ represents the feature vector and $Y_i$ represents the class label. Suppose follow a certain distribution $E_{(0,j)}(G_i)$, if $E_{(0,j)}(G_i) \neq E_{(j+1,\infty)}(G_i)$, then the group drift occurs at the timestamp $j + 1$, as shown in the Eq. (2):

$$\exists j : PEF_j(G_i) \neq PEF_{j+1}(G_i) \tag{2}$$

Group concept drift also refers to a set of related concept drift in the entire data stream. Group concept drift refers to the phenomenon of concept drift between different groups in a data set. In this case, the dataset can be divided into multiple groups, each of which may have a different feature distribution or category label distribution, resulting in conceptual inconsistencies between groups. Group concept drift may be caused by differences in the process of data collection or heterogeneity among different groups.

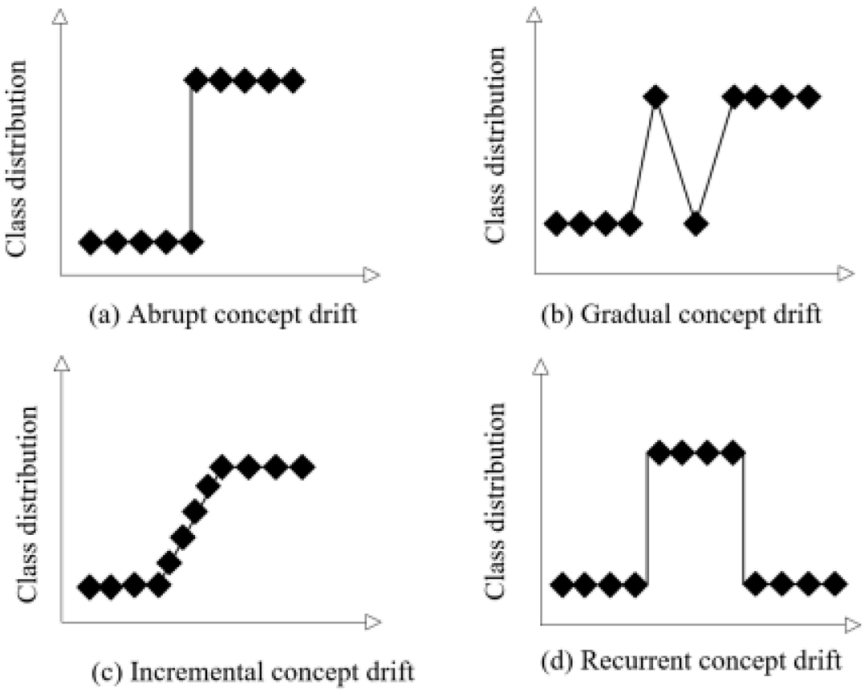



**Fig. 3.** Class by drift speed and duration.

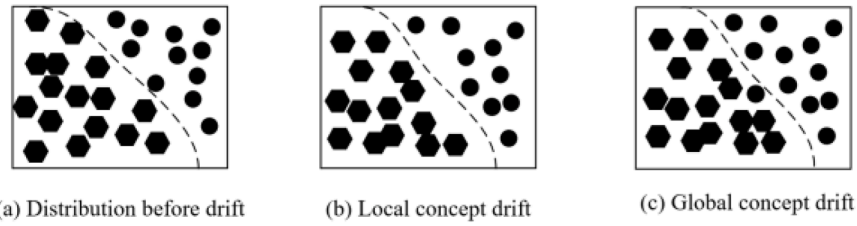



**Fig. 4.** Global and local concept drift.

Specifically, in group concept drift, the changes in the basic distribution of individual data streams not be regarded as drift, but these changes together represent the occurrence of group concept drift. Generally speaking, group concept drift refers to the phenomenon of conceptual inconsistency between different groups in the data set, so it is necessary to adopt targeted strategies to deal with the drift between different groups.

### 2.2. Concept drift processing method

The occurrence of concept drift will have a negative impact on the performance and stability of the supervised learning model. In order to deal with concept drift, researchers have proposed many processing methods, including statistics-based method [34], window-based method [35], data distribution-based method [20], model-based method and ensemble-based method [20]. In addition, after reading Reference [36], explanations were provided for the inclusion and exclusion criteria of the methods used in this paper. The method proposed in this paper mainly searches for relevant methods from several databases such as IEEE Xplore, ACM Digital Library, and Web of Science (WoS), and looks for related technologies by searching for "concept drift". Screen out the papers in the field of concept drift from the start of the research to 2025 from the above-mentioned database. Papers written in English and published in journals or conference proceedings are all likely to be included in this study.

The window-based method focuses on the occurrence of timestamps and events, and the window mechanism is used to process the interval data in the data stream during drift detection. Select the appropriate previous data to train the classifier and select the instances in the fixed or dynamic sliding window. Assuming that the old instance is not compatible with the new data classification, the useless old instance must be discarded. Therefore, the last batch of data and the last training example are used in the training. The method of fixing the size of the window is more effective when there is information about the time scale of the change. However, users need to weigh whether to choose a small window size that reflects the current distribution with rapid adaptability, or in a stable period when there is no conceptual drift, large window sizes that may improve accuracy and generalization if you have more instances [10]. The adaptive window method resizes the window to the length of the drift. Because the sample is usually maintained until the concept drifts, there is no need to select a priori unknown parameters in this case. The classical ADWIN [10] method judges concept drift based on window size adjustment and statistics index. Starting from the time window *win*, the window is dynamically enlarged when there is no obvious change in the upper and lower instances, and it is reduced when a change is detected. This method attempts to find the two child windows $win - win_1$ and *win* that display different averages, which means deleting the old part of the window $win_1$ that is different from the real data distribution. When the average error of the two windows exceeds the threshold $\sigma_{th}$ as shown in the Eq. (3), the drift occurs. Where is the observed variance of the instance in the window *win*, which *g* is the harmonic mean of the sum of the two child windows. The average expected confidence, $\omega' = \omega/m$ which represents the different possibilities in $m$ of the two sub-windows, can avoid the problem of multiple hypothesis testing, in which $m$ is the window length and $\omega$ is the expectation confidence.

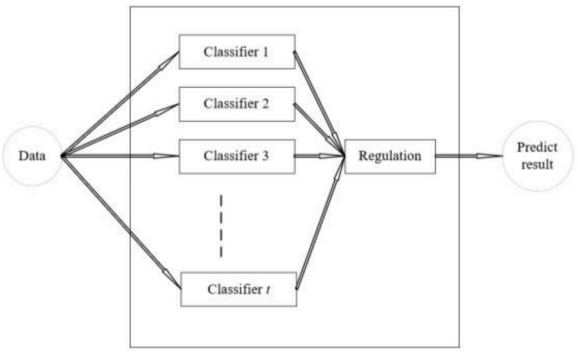



**Fig. 5.** The ensemble-based method.

$$\sigma_{th} = \sqrt{\frac{2}{g} \cdot v_{win}^2 \cdot \ln\frac{2}{\omega'} + \frac{2}{3g}\ln\frac{2}{\omega'}} \tag{3}$$

The ensemble-based method combines multiple base classifiers to define the final classification to improve the accuracy of the method. The classifier classifies specific recent instances into new classes or normal classes. The classifier should be adaptive because the data can change over time [35]. When the data distribution and its corresponding labels change, that is, concept drift occurs, any classifier trained on the currently available samples will become obsolete [3]. A set of dynamic classifiers can be used to observe performance or other metrics. If the performance degrades, the new classifier will replace the aging and poor performance classifier [4]. The ensemble-based method usually uses the weighted voting mechanism to combine the output of the learner to classify the examples, decompose the complex problems into multiple sub-problems, classify and predict the new incoming data instances and detect the concept drift at the same time. The prediction results are combined to get the final prediction results, as shown in Fig. 5. Develop an adaptive learning classifier ensemble for the data that may have concept drift. When the performance of one classifier becomes worse, replace the old classifier with other classifiers with better performance [4,5]. In the case of repeated concept drift, we can choose to retain or reuse the old classifier to save a lot of energy to retrain the new classifier. Dynamic ensemble selection of unbalanced data streams with conceptual drift (DES-ICD) [37] is a dynamic integrated classification framework that adapts to the change of drift concepts in class unbalanced data streams. The newly arrived data blocks are balanced by AnnSMOTE oversampling method, and then DES-ICD learns a new basic classifier on the balanced data block and combines it with the historical basic classifier to form a candidate classifier pool. For each query sample, DES-ICD constructs the performance of an optimal combination of candidate classifiers in its neighborhood. The DEnBoost square method based on Boosting ensemble [38] is a classifier ensemble method based on observation density and its corresponding label homogeneity metric. It uses data sampling density, local entropy and a simple ensemble construction method to detect local virtual concept drift.

Statistics-based methods compare historical and current data instance distribution through statistical tests, such as mean, hypothesis testing, regression, standard deviation and so on. In some cases, divergence is used to calculate the distance between distributions, such as KL divergence [39], JS divergence and K-S test. The most commonly used one is KL divergence, that is, relative entropy. The distribution distance is obtained by quantifying the difference between the two probability distributions. Suppose there are old data distribution An and new data distribution $B$, and the distance between the two distributions is shown in Eq. (4). Distance measurement is used to find the differences between data in different time ranges, and to measure the differences in data behavior over time. This method does not need tags and extra memory, and can quickly obtain the input characteristics of the model or output changing indicators, so as to detect concept drift earlier, and may lead to false positives if not handled correctly. Both the drift detection method DDM [7] and the early detection method EDDM [8] model the error as binomial variables $Y$ to calculate the expected error, and the expected error of the binomial distribution is expressed as $E(Y) = np_\lambda$, in which the expected error variance $\delta$ is expressed as $\sqrt{p_\lambda \cdot (1-p_\lambda)/n}$. DDM and EDDM separate use $p_\lambda + \delta_\lambda$and $\frac{p_\lambda + 2\delta_\lambda}{p_{ma}+2\delta_{ma}}$ to calculate the error values respectively, and a drift occurs when the specified threshold is exceeded.

$$KL(A \parallel B) = -\sum_i P(i) \cdot \log\frac{A(i)}{B(i)} \tag{4}$$

## 3. Handle single type

### 3.1. Real concept drift

Some of the existing concept drift processing methods are aimed at a specific type of concept drift, that is, can only deal with a single type of concept drift, and design methods for specific features of the type to deal with concept drift. According to the different types of concept drift, the processing methods of concept drift are classified from the point of view of dealing with real concept drift and virtual concept drift. Among them, real concept drift is more common, including abrupt concept drift, gradual concept drift, recurrent concept drift, and local concept drift. In addition to the method of dealing with real drift, the processing method of virtual concept drift is also explained.

#### 3.1.1. Abrupt concept drift

The concept drift of catastrophe usually lasts for a short time, but varies greatly. For the processing methods of drift data flow with abrupt concept, there are mainly statistics-based detection methods, integration-based methods, window-based detection methods and other methods. STEPD [9] use proportional statistical tests. The basic principle is to consider the recent accuracy and the overall accuracy. If the target concept is static, the accuracy of the classifier for the most recent instance will be equal to the overall accuracy starting from learning; and the recent significant reduction in accuracy indicates that the concept is changing. It is tested by calculating the statistics as shown in the Eq. (5), and its value is compared with the percentile of the standard normal distribution to obtain the observed level of significance $\widehat{H}$. $\alpha_0$ is the number of correct classifications in all $\beta_0$ instances except the most recent $W_c$ instances, $\alpha_1$ is the number of correct classifications in $W_c$ instances, $W_c = \beta_1$and $\widehat{h} = \frac{\alpha_0+\alpha_1}{\beta_0+\beta_1}$. If $H$ is less than the significance level $\widehat{H}$, the zero hypothesis $\frac{\alpha_0}{\beta_0} = \frac{\alpha_1}{\beta_1}$ is rejected and the alternative hypothesis $\frac{\alpha_0}{\beta_0} > \frac{\alpha_1}{\beta_1}$ is accepted, that is, the concept drift has been detected. STEPD uses $\lambda_a$ and $\lambda_b$ two saliency levels, stores the instance in short-term memory when $H<\lambda_a$ satisfied, then rebuilds the classifier from the stored

instance, and resets all variables if $H<\lambda_b$. When $\beta_0+\beta_1 \geq 2W_c$ is satisfied, the drift is detected, and if $H\geq\lambda_a$, the stored instance is deleted. Although this method can detect concept drift accurately and quickly, it will also has the possibility of false detections. The classical drift detection method DDM [7] is based on the online error rate and uses two confidence levels. If the alert level is exceeded, the instance is stored to predict possible changes in the context. If the drift level is exceeded, the concept drift is considered to have occurred, the model is reset and the instance stored since the warning level is triggered is used to learn the new model. This method is efficient and simple, but its performance will be affected when the concept is very long. The multiscale drift detection test MDDT [40] is based on a resampling scheme and a paired student *t*-test. The eigenvalue fluctuation is mainly detected when locating the abrupt drift point, and the detection program is applied in a wide and narrow range. The multi-scale structure not only reduces a large number of time constant checking process, but also filters the noise in the detection function. This method has low computational cost and high average accuracy, but it does not adapt to some classifiers.

$$T(\alpha_0,\alpha_1,\beta_0,\beta_1)=\frac{\left|\frac{\alpha_0}{\beta_0}-\frac{\alpha_1}{\beta_1}\right|-0.5\left(\frac{1}{\beta_0}+\frac{1}{\beta_1}\right)}{\sqrt{\widehat{h}(1-\widehat{h})\left(\frac{1}{\beta_0}+\frac{1}{\beta_1}\right)}} \tag{5}$$

In addition to statistics-based methods, there are a large number of window-based processing methods. For example, ADWIN [10] method increases the window when there is no obvious change, and vice versa. For drift detection, ADWIN checks two sub-windows $w_0$ of size $\alpha_1$ and $w_1$ size $\alpha_2$ by sliding the window over the prediction results, where the size of the window $w$ is $k=\alpha_1+\alpha_2$, the harmonic average $n=1/\left(\frac{1}{\alpha_1}+\frac{1}{\alpha_2}\right)$, $\theta'=\theta/k$ is the confidence level. Therefore, when the average value of the two child windows is quite different, a drift occurs when it exceeds the threshold shown in Eq. (6), that is, a concept drift occurs when $\mu w_0-\mu w_1\geq\xi$. The McDiarmid drift detection method (MDDM) [41] uses a sliding window to detect the prediction result and associates the window entry with the weight. When dealing with an example, the McDiarmid inequality is used to determine whether the concept is drifting. This method has shorter detection delay and lower false negative rate, while maintaining high classification accuracy. The multistage weighted drift detection method (MWDDM) [42] proposes two variants of MWDDM_H and MWDDM_M based on Hoeffding inequality and McDiarmid inequality respectively. In this method, a double sliding window mechanism is introduced to assign a larger weight to the new instance, otherwise a smaller weight is assigned, and the weight difference is smaller when the "stable level" is reached, and becomes larger when the "warning level" is reached. At the drift level, the boundary values generated by the two inequalities are used to determine whether the difference between the correctly predicted weighted average and the maximum weighted average exceeds the predefined threshold to report whether the drift occurs. If drift occurs, the classifier is reset and retrained. The two variants can maintain high classification accuracy in the case of low false detection rate and low missed detection rate, but lack robustness to noise.

$$\xi=\sqrt{\frac{1}{2n}\ln\left(\frac{4}{\theta'}\right)} \tag{6}$$

For data streams with abrupt drift, there is a classifier integration method in addition to a single detection method. AWOE [43] is an integration method with internal detectors. This method combines the advantages of block-based weighting and on-line processing, uses an adaptive window for change detection, dynamically adjusts the size of the data block, and replaces the poor classifier once the change is detected. The sliding window is divided into two equal-length sub-windows $Win_{left}$ and $Win_{right}$, the relative entropy is dynamically obtained from the window by using the transformation detector, that is, the spatial primitive of the event is summed, as shown in Eq. (7). Using relative entropy as a measure, the difference between the two sub-windows is compared with the Hoeffding boundary to determine whether the target concept is drifting. When the relative entropy is greater than the threshold, the change is detected and the window is deleted. ADOB [15] aims to speed up the concept drift after the classifier is restored. This method has high accuracy and regularity, but there is no significant change in time and space efficiency. BOLE [44] is improved on the basis of ADOB by modifying the voting strategy and replacing the drift detector to improve the classification accuracy. However, BOLE also has some drawbacks, such as giving up some theoretical guarantees and parameter settings have a significant impact on the detection results. In addition to the above common methods, there are other processing methods. Diversity Measure As a New Drift Detection Method (DMDDM) [45] based on the ever-changing incoming data, inconsistency measures $D_{\alpha,\beta}=X^{10}+X^{01}$ are used to calculate the diversity between the base classifier $\alpha$ and $\beta$ to observe the various responses of the classifier to convection changes. Wherein, $X^{10}$ represents the number of instances of $C_v$ prediction category 1 and $C_w$ prediction category 0, and $X^{01}$ indicates the number of instances of $C_w$ prediction category 1 and $C_v$ prediction category 0.

$$KL\left(Win_{left}\middle|Win_{rifht}\right)=\sum_{a\in A}P_{Win_{left}}(a)\cdot\log\frac{P_{Win_{left}}(a)}{P_{Win_{right}}(a)} \tag{7}$$

### 3.1.2. Gradual concept drift

The drift duration of gradual concept drift is long, and the speed of change is relatively slow. In view of this feature, the methods to solve the gradual concept drift include statistics-based method, window-based method, ensemble-based method and so on.

Among the statistics-based methods, the classical method EDDM [8] is based on DDM. The square method detects the change by the number of instances between two classification errors. At the same time, the threshold is set for detection in the case of "warning level"

and "drift level" respectively, as shown in Eq. (8) and Eq. (9). Among them, the average distance between the two errors is *Pi*, the standard deviation between the two errors is *Si*, $\alpha$and $\beta$are indicates the threshold, and the value of $P_{MAX} + 2 \cdot S_{MAX}$ corresponds to the point with the largest distance distribution between the errors. Based on DDM and other heuristic modifications, RDDM [46] solves the problem of performance loss in the face of very long concepts by periodically shortening the number of instances of ultra-long stability concepts. Discarding the old long concepts shortens the drift time and improves the accuracy. At the same time, due to the abandonment of some concepts, the accuracy rate in the big data set is not high.

$$(Pi + 2 \cdot Si)/(P_{MAX} + 2 \cdot S_{MAX}) < \alpha \tag{8}$$

$$(Pi + 2 \cdot Si)/(P_{MAX} + 2 \cdot S_{MAX}) < \beta \tag{9}$$

The ensemble-based approach includes real-time adaptive classifier (Adaptive JIT) [47], which integrates change detection tests to identify distribution changes in the data generation process and to delete outdated training samples. This method allows the classifier to automatically improve the accuracy under stable conditions and respond to changes under unstable conditions in a timely manner. A novel change detection test whose conditional probability distribution expectations are shown in Eq. (10) and Eq. (11), follow the process of a polynomial trend, and when this trend changes and other statistical properties of the *j-th* process change. Where time $s \in J_j$, $g_{1j}$ and $g_{2j}$ are the average values of $\sigma_1$ and $\sigma_2$ under stationary conditions, the parameter description of $f_{\Phi_j}(s)$ is given by $\left\{\left(\Phi_j, U_j\right)\right\}$, $\Phi_j$ is a set of coefficients of the polynomial formula $f_{\Phi_j}(s)$ defined on the *j-th* interval, and *M* is the number of intervals. Therefore, the process *Y(s)* of producing observation in time becomes as shown in the Eq. (12). Among them, under the stationary conditions of $E\left[\theta_{1j}\right] = g_{1j}$ and $E\left[\theta_{2j}\right] = g_{2j}$ are exist, $\theta_{1j}$ and $\theta_{2j}$ are the probability density functions respectively characterizing the distribution of $\sigma_1$ and $\sigma_2$. Using window method Fuzzy window concept Drift Adaptive method (FW-DA) [48] introduces fuzzy sets to describe the concepts before and after drift, and uses three fuzzy sets, old concepts, new concepts and noise, to describe the relationship between event correlation distribution. And without considering that the weights of all data instances are equal, but calculate the test statistics according to the data instances weighted by window correlation membership degree. In addition, there is an enhancement method combined with Hoeffding adaptive tree (EnHAT) [49], which uses Hoeffding adaptive tree and J48 decision tree to build an updated model. When the Hoeffding adaptive tree updates its model through a single recently arrived data window, new data blocks may enter the system. The Visual drift detection (VDD) [50] is also a method to deal with gradual drift, a visualization method based on drift graph and drift graph, which is used for the classification of process drift. This method declares the clustering of behavioral constraints, which exhibit similar changing trends over time and automatically detects the changes in drift points. For each step of these analyses, different visualizations and various measures indicating the type of drift are provided.

$$E\left[P_j(Y|\sigma_1, s)\right] = f_{\Phi_j}(s) + g_{1j} \tag{10}$$

$$E\left[P_j(Y|\sigma_2, s)\right] = f_{\Phi_j}(s) + g_{2j} \tag{11}$$

$$Y(s) = \begin{cases} f_{\Phi_j}(s) + \theta_{1j}, x(s) = \sigma_1 \\ f_{\Phi_j}(s) + \theta_{2j}, x(s) = \sigma_2 \end{cases} \tag{12}$$

### 3.1.3. Recurrent concept drift

Recurrent concept drift may be caused by periodic external factors or environmental changes, and the duration is relatively fixed. The methods to deal with data streams with recurrent concept drift include metamodel-based method, ensemble-based method and clustering-based method.

The conceptual analysis framework Concept Profiling Framework (CPF) [51], which deals with recurrent concept drift, is a meta-learner, which mainly classifies data streams with recurrent drift. The framework can use the observed model behavior to accurately identify recurrent concepts over time without relying on additional information about the underlying concepts. In order to avoid the risk that the classifier set may continue to grow, resulting in increasing performance overhead, a decay mechanism is used to limit the size of the classifier set. The enhanced Conceptual Contour Framework Enhanced Concept Profiling Framework (ECPF) [52] is based on CPF, which aims to identify recurring concepts and reuse previously trained classifiers so as to achieve accurate classification immediately after drift. The advantage of ECPF lies in how it uses the classification similarity of the new data to quickly identify the best classifier for reuse between the new classifier and the existing classifier. It always trains a new classifier and a reused classifier and retains a more accurate classifier when concept drift occurs. Finally, it creates a copy of a reusable classifier, so a classifier that is well suited for recurring concepts will not be affected by the training of different concepts. When comparing classifier $c_1$ with classifier $c_2$, the score of each instance is calculated by $Scor(x, c_1, c_2) = \begin{cases} 1, c_1(x) = c_2(x) \\ 0, c_1(x) \neq c_2(x) \end{cases}$, where $c_1(x)$ and $c_2(x)$ respectively are the classification error of $c_1$ and $c_2$ on a given instance *x*. Then, calculate the similarity of the two classifiers on a series of instances, using the example formula of length seen during the warning drift, as shown in Eq. (13).

$$Sim(X_1, \ldots, X_m, c_1, c_2) = \frac{\sum_{j=1}^{m} Scor\left(X_j, c_1, c_2\right)}{m} \tag{13}$$

The ensemble-based method includes a recursive dynamic weighted majority voting system (Recurring Dynamic Weighted

Majority system, RDWM) [53], which includes a primary online set and a secondary set integration system, which is mainly used to deal with repeated drift. Among them, the primary online set represents the current concept and will be trained and updated, and the secondary set represents the old concepts since the beginning of learning, including the most effective classifiers copied from the primary set. In addition, there is also a method for identifying recurrent concepts in data stream clustering [54]. During the recurrent of the outline, the method retrieves the best matching model from the repository and reuses it. This method uses a detector based on sequential analysis, namely Page-Hinkley test, to detect the parameters that represent the change, that is, the average distance between the data point and its nearest cluster center. Let $D_a$ represent the distance variable monitored by the Page-Hinkley test, that is, the distance between the data point $a$ and the nearest clustering center in the current data partition. The cumulative variable is calculated as shown in Eq. (14). Where $\overline{D}_a = \sum_{b=1}^{a} D_b / a$, $m_t$ is the sum of the deviation between $D_a$ and its average up to the time $t$, $\varphi$ indicating the tolerance level, the threshold $\mu_a$ and $\mu_b$ respectively represent the alarm threshold and the warning threshold, and $\mu_a > \mu_b$. A warning is issued when the difference $Min(m_a, a = 1, \ldots t)$ between $m_t$and its minimum value $M_t$, is greater than $\mu_b$, that is, $m_t - M_t > \mu_b$. when the difference value is greater than $\mu_a$, the alarm is triggered.

$$m_t = \sum_{a=1}^{t} (D_a - \overline{D}_a - \varphi) \tag{14}$$

### 3.1.4. Local concept drift

In addition to dealing with the relevant drift types described above, there are also processing methods for solving data streams with local drift. Local concept drift is that only some samples are affected by the drift, while other samples remain stable, which is usually caused by changes in specific conditions or events. Based on this feature, the method based on region density is generally used to deal with local concept drift.

For data streams with local drift, the density synchronous drift adaptive method (Density Synchronized Drift Adaptation, LDD-DSDA) [55] uses local drift to measure the possibility of local drift, which can continuously monitor the change of regional density. Instead of pausing all historical data after drift, this method synchronizes the regional density difference according to the degree of local drift. LDD-DSDA merges the existing data blocks $D_{tra}$ with the most recently buffered data blocks $D_{buf}$, and resamples a number of new training data blocks $D_{tra}$, so that the new training data blocks $D_{tra}$ have the same distribution as the training data blocks $D_{buf}$. In order to achieve this goal, the following data merging rules are proposed: if no density difference is detected, that is, $\frac{|D_1^{de}|}{|D_1|} = \frac{|D_2^{in}|}{|D_2|} = 0$, $D_{tra} \cup D_{buf}$ is then returned, and if there is a regional density drift, the data examples of the drift are merged as shown in the Eq. (15). There is another method based on region density estimation: nearest neighbor density variation recognition Identification (NN-DVI) [56]. The method consists of three parts, namely, a space partition model based on k-nearest neighbor, which converts unmeasurable discrete data instances into a group of shared subspaces for density estimation. the distance function of the density difference in the cumulative subspace and quantifies the overall difference and the statistical significance test of the confidence interval which can determine the concept drift. The distance used in NN-DVI follows normal distribution and is sensitive to local drift. In addition, the weighted integration of different cases (Diverse Instance-Weighting Ensemble, DiwE) [16] based on weighted integration can also solve the problem of local concept drift in data flow classification. This square method is based on the diversity measurement method of whether the set members agree with the change probability of regional distribution, regards the change of regional distribution as case weight, and designs different schemes to construct different regional sets so as to get different drift estimation results. to adapt to the existence of diversity. Based on dynamic ensemble selection, a DES-ICD [37] method for unbalanced data streams with concept drift is proposed. Aiming at block-by-block data, this method proposes a synthetic minority oversampling technique (SMOTE with adaptive nearest neighbors, AnnSMOTE) with adaptive nearest neighbors to generate new instances that conform to the new concept. A base classifier is then created on the newly arrived data blocks balanced by AnnSMOTE and combined with the historical base classifier to form a candidate classifier pool. When the number of base classifiers in the pool is greater than the maximum, the old base classifiers are discarded.

$$\begin{cases} D^{in} = D_{tra}^{in} \cup D_{buf}^{de} \\ D^{st} = D_{tra}^{st} \cup D_{buf}^{st} \\ D^{de} = D_{tra}^{de} \cup D_{buf}^{in} \end{cases} \tag{15}$$

## 3.2. Virtual concept drift

Virtual concept drift is not due to the change of the concept of data generation itself, but to simulate the drift phenomenon by manipulating the data or introducing noise. For virtual concept drift, there are model-based method, ensemble-based method and other concept drift processing methods.

### 3.2.1. Model-based method

The incremental data stream learning method based on Gaussian mixture model IGMM with support to Concept Drifts (IGMM-CD) [57] abandons outdated components and controls the maximum number of components, so it is much faster than the original incremental algorithm. Although this method has good efficiency, it is highly dependent on the parameter $T$ which controls the number of components of each class. On this basis, an improved on-line Gaussian hybrid model and noise filter is proposed to deal with virtual

and real concept drift On-line Gaussian Mixture Model with Noise Filter for Handling Virtual and Real Concept Drifts (OGMMF-VRD) [58]. In order to overcome the influence of noise, the *k*-Disagreeing Neighbors (*k*DN) definition equation is used as the noise filter as shown in Eq. (16). Consider the need to determine whether an observation is noise. *k*DN indicates that the scores of similar *y* are not shared among the *k* nearest neighbors of the observation. A value close to 0 indicates that it is easy to classify and is unlikely to be noise, while a value close to 1 indicates that it is difficult to classify and may be noise.

$$kDN(a) = \frac{|\forall a' | a' \in kNN(a) \wedge label(a')|}{k} \tag{16}$$

### 3.2.2. Ensemble-based method

Among the methods based on integration, the classic one is based on the density of observations and the uniformity measure of corresponding tags to construct classifier integration, that is, dynamic environment promotion (Dynamic Environment Boosting, DEnBoost) [24] method. This method allows new data points to be added incrementally to the model without a complete retraining process. Because the information density metric allows it to detect local new information, the problem of virtual concept drift is solved successfully. The distribution of the observations and their corresponding labels defines the focus area of the subsequent classifier, and the integrator uses its local ability to combine. The initial dataset is $B$, and the information density difference between the dataset $B(a)$ and $B(a+b)$ is $\Delta_{B(a),B(a+b)}(y)$, assume that $\phi$ is a symmetric function. The density change can be measured using a common radius in two data sets, as shown in Eq. (17). An incremental learning and model selection method based on neural network ensemble is proposed for virtual concept drift environment [59]. In this method, a statistical model of virtual concept drift is constructed, and the model selection criterion of radial basis function neural network in this environment is established by using the environment model. The integrated self-structured divine sutra network (Ensemble Self-Structuring Neural Network, ESSNN) [60] framework has the ability to accommodate new knowledge without discarding previously learned knowledge. Once a new dataset block is collected, the integration is evaluated. If the overall accuracy of the combination results from the integration decreases significantly, a new classifier is created using a self-structured neural network and added to the integration. ESSNN simplifies the structure of neural network classifier with the least user intervention.

$$\Delta^{c}_{B(a),B(a+b)}(y) = \phi\left(\beta_i^{(a+b)}(y,c) - \beta_i^{(a)}(y,c)\right) \tag{17}$$

### 3.2.3. Other processing methods

In addition to the above model-based and various integration methods, there are many other methods. The updated version of the drift test (Dynamic Selection Based Drift Handler, Dynse) [61] framework based on dynamic selection adds a pruning module, that is, a module that prevents the pool size from increasing infinitely. The virtual drift detection method (VDDM) [62] uses multivariate nonparametric statistical tests to compare all attributes to identify when concept drift occurs. An effective framework for estimating the generalization error of regression function [63], the framework does not pay attention to the change of the distribution of instances, but detects the change of regression estimation error, and drift occurs if the error is greater than the threshold. For example, an unsupervised concept drift detection method for detecting virtual drift in non-stationary data [64]. The *K*-means clustering method is applied to the related features to find the virtual drift of the flow. This method analyzes the virtual drift of data flow by considering the changes of data distribution of recent and current window data instances. The data distribution boundary remains unchanged, but when the cluster data distribution changes, that is, when the centroid or farthest point or intercommunity distance changes, the concept drift is detected as shown in Eq. (18). Where the centroid distance between the nearest window $D_x$ and the current window $D_y$ is $dis(D_x, D_y) = \sqrt{(D_{x1} - D_{y1})^2 - (D_{x2} - D_{y2})^2}$, the farthest point of the cluster is $FL$, and $D_{xj}$ is the cluster centroid of the nearest window $D_x$.

$$D_{xj} \neq D_{yj} \cup FL_j \neq FL_{j+1} \cup dis_j(D_x, D_y) \neq dis_{j+1}(D_x, D_y) \tag{18}$$

## 3.3. Analysis and summary

This section analyzes and summarizes the methods mentioned in Sections 3.1 and 3.2. Firstly, the method was systematically summarized and generalized from aspects such as the type of drift applicable to the method, core ideas, experimental datasets, comparison methods, as well as advantages and disadvantages. In addition, methods such as MDDM, DDM, ADWIN, CUSUM, HDDM, FHDDM, EDDM, Page-Hinkley and RDDM were selected for experimental comparison. Based on the two base classifiers, Hoeffding tree and Naive Bayes, compare their classification accuracy rates on abrupt change datasets and gradual change datasets, thereby intuitively observing the drift types applicable to different methods.

The methods for dealing with abrupt drift data streams include statistics-based DDM, STEPD, MDDT methods, window-based ADWIN, MDDM, MWDDM methods, integrated ADOB, AWOE, BOLE and DMDDM methods, statistics-based EDDM and RDDM methods, integrated Adaptive JIT methods, window-based FW-DA, EnHAT methods and process drift VDD methods. There are CPF, ECPF and RDWM methods based on metamodel to deal with repeated drift data streams, LDD-DSDA, NN-DVI, DES-ICD and DiwE methods based on region density to deal with local drift data streams, DEnBoost and Dynse methods based on integration, OGMMFVRD and IGMM-CD methods based on Gaussian mixture model, VDDM based on statistics, regression-based methods and ESSNN based on neural networks. A detailed summary of the drift types, usage techniques, datasets, comparison methods, and advantages and disadvantages of the methods in this chapter is shown in Table 1.

In the study of reference [42], the synthetic data streams Mixed and Sine1 with abrupt drift and the data streams LED and Circles with gradual drift are tested by using Hoeffding tree and Naive Bayesian classifier, respectively, and averaged here. The experimental results are shown in Fig. 6. In this experiment, in the gradual data stream, ADWIN based on Hoeffding tree and Naive Bayes has an average accuracy of 78.94 % and 78.43 % respectively, which is the only method in the experiment that is lower than 80 %. The second is SeqDrift2, which has the accuracy of 82.99 % and 80.34 % respectively. FHDDM, HDDM and CUSUM have an average accuracy of 87.465 %, 87.445 % and 87.435 % in gradient data streams, whether using Hoeffding tree as base classifier or Naive Bayes as base classifier. Among them, the average accuracy of FHDDM with better effect is 1.11 times higher than that of ADWIN with poor effect. In the abrupt data stream, EDDM based on Hoeffding tree and Naive Bayesian classifier has an average accuracy of 82.68 % and 81.87 % respectively, while the classical method DDM has an average accuracy of 83.92 % and 81.34 %, respectively. In the case of non-discrimination base classifier, the average accuracy of MDDM, CUSUM, Page-Hinkley, RDDM, ADWIN, SeqDrift2, HDDM and FHDDM is 84.97 %, 84.86 %, 84.4 %, 84.79 %, 84.79 %, 84.49 %, 84.61 % and 84.98 %, respectively. The average accuracy of MDDM with better effect is 1.033 times higher than that of EDDM. The experimental results show that both ADWIN and SeqDrift2 have good average accuracy in abrupt data streams, regardless of whether they are based on Hoeffding tree or Naive Bayes, but they do not perform well in gradual data streams. On the contrary, in the gradual data stream, whether it is based on Hoeffding tree or Naive Bayes, the average accuracy of DDM, EDDM, HDDM and FHDDM is higher. However, DDM and EDDM are not effective in abrupt data streams.

## 4. Handle multiple type and group type

In a single data stream, there is not just one type of concept drift; there may also be other types of concept drift. Most of the two types of management are dealing with abrupt and gradual change, abrupt and recurrent, abrupt and increment, real and virtual. There are abrupt type, gradual type and incremental type, abrupt type, gradual type and recurrent type, abrupt type, gradual type, incremental type and recurrent type. In multi-data streams, there may be multiple related types of concept drift, namely group concept drift [3]. This section describes in detail the concept drift detection methods that deal with these types.

### 4.1. Multiple type

#### 4.1.1. Abrupt and gradual concept drift

For data streams with abrupt drift and gradual drift, statistics-based, data distribution-based, window-based and ensemble-based methods can be adopted.

Window-based method Fast Hoeffding drift detection method (FHDDM) [65] uses sliding window and Hoeffding inequality to detect drift. Suppose the difference between the maximum probability and the nearest probability of the correct detection in $m$ sequence $\Delta p = p'_{\max} - p'_{near} \geq 0$, and the allowable error probability $\rho$ is given, if $\lambda_a = \sqrt{\frac{1}{2n}\ln\frac{1}{\rho}}$ as a drift occurs. Unsupervised concept drift detection method (One-Class Drift Detector, OCDD) [66] uses a single-class learner with a sliding window to detect concept drift. Drift occurs when the percentage of outliers $\beta = \sum_{a=1}^{c} D[a]/c$ is greater than the threshold $\varepsilon$, otherwise there is no drift. Among them, $W$ stores the latest data, $D$ stores the prediction data of a single classifier of size $c$, and $|W| = |D| = c$. The conceptual drift recognition method based on multiple sliding windows (Concept Drift Type Identification Method is proposed based on Multi-sliding Windows, CDT_MSW) [11] can first detect the drift position, and then calculate the drift length by analyzing the stability of the data distribution. A single basic sliding window is used to detect changes in data distribution and drift position. During the detection process, the basic window $win_A$ and $win_B$ of size $s$ is constructed and is located in the same position at the initial time. The accuracy of the classifier trained by the data in window $W$ is called the real-time accuracy of $W$. When entering the data and filling the $win_A$ and $win_B$, the real-time accuracy $\beta'_A$ and $\beta^0_B$ of $win_A$ and $win_B$ is obtained respectively. Because the basic window coincides at the initial time, $\beta'_A = \beta^0_B$. As a reference, it is static in the whole testing process. When $win_B$ sliding forward with step $s$, the new data are filled in $win_B$ in turn, and the corresponding real-time accuracy of $win_B$ is obtained. Then, the detection flow ratio is calculated according to the real-time accuracy of $win_A$ and $win_B$. Assuming that the real-time accuracy at position I is at B, the flow ratio $\overline{P}^i_{\text{det}}$ can be calculated as shown in Eq. (19). When $\overline{P}^i_{\text{det}}$ satisfies the inequality $\overline{P}^i_{\text{det}} < v$, where $v$ is the drift position detection parameter, the real-time accuracy of $win_B$ decreases obviously, indicating that there is a big difference in the data distribution between $win_A$ and $win_B$. Therefore, when the inequality is satisfied, there may be a concept drift in the data in $win_B$, and the position $i$ is regarded as the drift position.

$$\overline{P}^i_{\text{det}} = \beta^0_B / \beta'_A \tag{19}$$

The statistical method Wilcoxon rank sum test drift detector (Wilcoxon Rank Sum Test Drift Detector, WSTD) [67] changes the statistical tests used to issue warnings and drift signals, and limits the size of the old STEPD window. In addition, the effective implementation of rank sum test is also provided. The $n_1 + n_2$ observations of the two samples are merged and sorted in ascending order to get the rank of each observation in the interval $[1, n_1 + n_2]$. In the case of the same observation, their ranks are replaced by the average of their ranks in distinguishable cases. In addition, the rank sum of the two samples is calculated and the smallest $R$ is selected. The test statistics are calculated by the generalized equation as shown in the Eq. (20). Among them, $\alpha_R = a_1 \times (a_1 + a_2 + 1)/2$ and $\beta_R = \sqrt{a_1 \times a_2 \times (a_1 + a_2 + 1)/12}$ respectively are the overall average and standard deviation, $a_1$ and $a_2$ respectively represents the minimum and maximum sample size. Then, when the obtained $z$ value is in the rejection region, the zero hypothesis that the two samples have the same distribution is rejected, and the accepted samples come from different distributions. Another possibility is to

**Table 1**
Handle a single type of method summary.

| Algorithm | Suitable for | Method | Data set | Contrast algorithm | Advantages and disadvantages |
|---|---|---|---|---|---|
| DDM[8] | abrupt drift | error rate | Mixed,Guass,Circles etc. | Decision tree,Neural network,Perceptron | It has high computational efficiency and can be integrated with other methods, but it is not effective on real data sets |
| STEPD[10] | abrupt drift | statistical test | Mixed,Guass,Circles etc. | DDM,EDDM,ACED | Can quickly and accurately detect concept drift and the existence of false detection |
| ADWIN[11] | abrupt drift | sliding window | ELEC | – | Low memory and time, can't solve many kinds of problems |
| ADOB[15] | abrupt drift | online Boosting | Sine,Stagger,LED,RBF,Poker etc. | DDD,DWM,AUE2, AdwinBag, Leveraging etc. | With high accuracy, its performance can be further improved through optimization methods |
| MDDT[40] | abrupt drift | *t*-test, resampling | SEA,RBF,Sine,Stagger etc. | HCDT,KNNWA, CUSUM, Page–Hinkley | The calculation time is short and can't adapt to some classifiers |
| MDDM[41] | abrupt drift | Sliding window, McDiarmid inequation | Mixed,Sine,LED,Circles etc. | DDM,EDDM,RDDM, FHDDM etc. | The detection was rapid and the false negative rate was small |
| MWDDM[42] | abrupt drift | multilevel weighting, double sliding window | Mixed,LED,Sine etc. | BDDM,EDDM,DDM etc. | Low false positives and false negatives, fast detection drift, lack of robustness to noise |
| AWOE[43] | abrupt drift | online Boosting, adaptive | Poker,Hyperplane,LED etc. | DDM,EDDM,ADWIN etc. | Maintain high stable accuracy and cannot adapt to labeled data |
| BOLE[44] | abrupt drift | online Boosting, adaptive | Poker,Stagger,Waveform etc. | DDD,DWM,LEVBAG etc. | High accuracy, sensitive to concept drift, there will be false alarms |
| DMDDM[45] | abrupt drift | diversity measure | Mixed,SEA,Sine etc. | FHDDM,DDM, ADWIN etc. | High temporal and spatial efficiency, fast detection, can't solve many kinds of problems |
| EDDM[9] | gradual drift | estimated distribution | Mixed,Sine,Gauss etc. | DDM | It is more sensitive when there is noise, and the parameters lack an adaptive mechanism |
| RDDM[46] | gradual drift | reactive | Mixed,Sine,LED etc. | DDM,STEPD etc. | There is a better balance between true positive tests and false positive tests |
| Adaptive JIT [47] | gradual drift | adaptive, JIT model | Application D1,Application D2 | – | The detection of gradient drift is good in non-stationary environment |
| FW-DA[48] | gradual drift | fuzzy set theory | SEA,ELEC,Airline etc. | DDM,ECDD,HDDM etc. | Missing membership function for noise window |
| EnHAT[49] | gradual drift | decision tree, sliding window | Poker,ELEC,SEA,Airlines etc. | HAT | The computational complexity is low, and the accuracy has no obvious difference compared with other methods |
| VDD [50] | gradual drift | process drift detection | ConditionalMove, ConditionalToSequence,Loop etc. | – | Providing different visualizations, irregular sampling data can affect the analysis results |
| CPF[51] | Recurrent drift | metamodel | ELEC,Airlines ,Poker etc. | RCD | Fast and memory efficient |
| ECPF[52] | Recurrent drift | metamodel | LED,Stagger,Poker etc. | ARF,AUE,CPF etc. | It has good effect on recurrent drift detection |
| RDWM[53] | Recurrent drift | ensemble-based method | Stagger,Hyperplane,Power supply etc. | EDDM,DWM | High generalization accuracy |
| Namitha et al. 2020[54] | Recurrent drift | cluster | KDD-CUP'99,Weather,Synthetic etc. | – | Simple and effective, all models in the repository need to be checked |
| LDD-DSDA [55] | local drift | nearest neighbor distribution | Elec,Weather,Spam Filtering | ECDD,AUE,HAT etc. | It performs well in detecting local drift |
| NN-DVI[56] | local drift | area density estimation | Mixed,Circles,Poker etc. | MMD,KL,CM etc. | Can effectively detect local drift, performance is limited by the window size |
| DiwE[16] | local drift | ensemble-based method | SEA,LED,RBF etc. | ADWIN-ARF , LevBagkNN , OnlineAUE etc. | Incremental drift may cause a drift warning to be triggered |
| Yamauchi et al. 2010[57] | virtual drift | incremental learning | Concrete compressive strength, auto-mpg ,cpu-performance etc. | Org-RBFNN, WRBFNN | Incremental learning environments that produce appropriate model selection criteria |
| ESSNN[58] | virtual drift | self-structured neural networks | 1,2,3 | BN,SVM,LR,RIPPER etc. | It has good stability and plasticity, and can't solve many kinds of problems |
| IGMM-CD [59] | virtual drift | based on GMM classifier | 1CDT,4CR,5CVT,1CHT etc. | IGMM | The method is fast, efficient and eliminates Gaussian component |
| OGMMF-VRD [60] | virtual drift | online Gaussian mixture model | Sine2,Circles,NOAA,PAKDD etc. | OGMMF-VRD,GMM-VRD,Dynse | To achieve better robust noise, it does not perform well in class unbalanced data streams |

(*continued on next page*)

**Table 1** (*continued*)

| Algorithm | Suitable for | Method | Data set | Contrast algorithm | Advantages and disadvantages |
|---|---|---|---|---|---|
| DEnBoost[38] | virtual drift | ensemble-based method | RBF | – | It does not depend on the classifier architecture chosen and does not require much parameter adjustment |
| Dynse[61] | virtual drift | bayes classifier | SEA,Guass,For,Ckr etc. | AUE,AWE,ADACC, DDM etc. | Add pruning strategy without having to adjust parameters too much |
| VDDM[62] | virtual drift | statistical test | Circles12,Car,Credit-a,Vote etc. | KS,TT,WRS | High detection accuracy, the window is a fixed window |
| Agrahari et al. 2022[63] | virtual drift | data distribution | LED,Sine,Spam,Airlines etc. | ADWIN,ECDD,SEED, STEPD etc. | It can effectively deal with high-dimensional data, but cannot effectively deal with noise and outliers |
| Oikarinen et al. 2021[64] | virtual drift | regression model | Air quality,Flight delay,Bike rental etc. | – | The parameters need to be optimized |

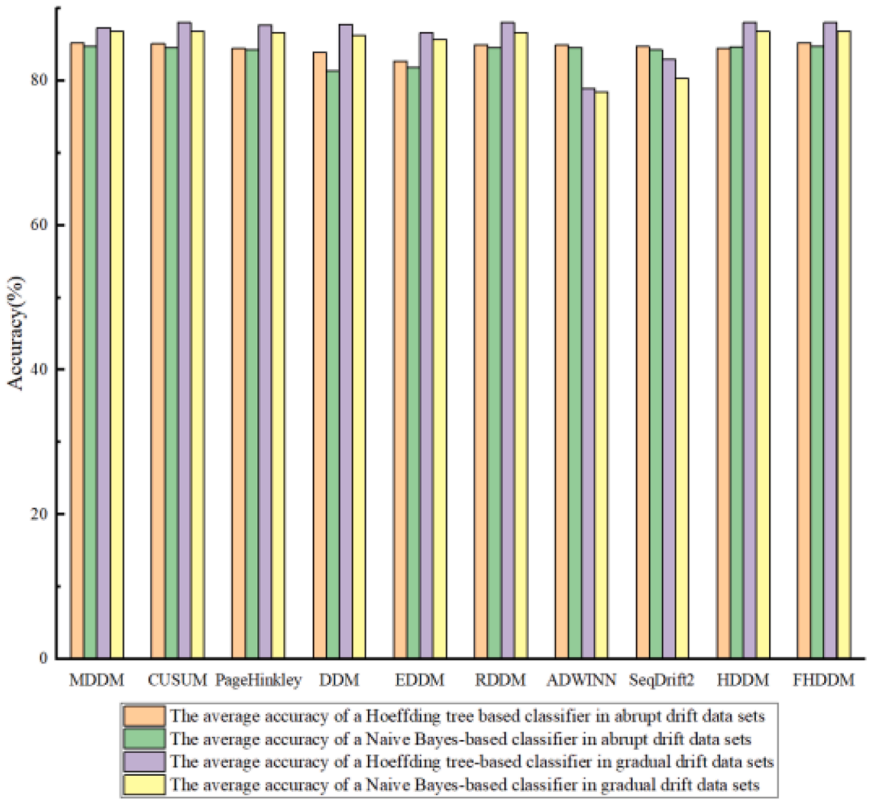



**Fig. 6.** The average accuracy is obtained by Hoeffding tree and Naive Bayes based classifiers in abrupt and gradual data streams.

determine the strength of the zero hypothesis by calculating the *p*-value and estimating the strength of the corresponding rejection evidence of whether the two samples have the same distribution. On the basis of proving the learning boundary of unsupervised concept drift detection in data stream based on the method stability framework, the Plover [68] method is designed. According to the basic characteristics of the corresponding application field, each data stream may need different measurement functions to identify concept drift. This method uses different measurement functions to detect drift, and the measurement functions include statistical moment and power spectrum, so that statistical moment is used as the measurement function to support the detection of drift from the changes of mean, variance, skewness and kurtosis. Power spectrum analysis is used as a measuring function to detect the drift of data amplitude and frequency. The measurement function plays an important role in detecting changes, and the selection of parameters will also affect the detection results. In addition, the statistical drift detection method (Statistical Drift Detection Method, SDDM) [14], as a statistical drift detection method adapting to the input flow characteristics, can detect the change of drift monitoring data distribution without feedback to the output of the classifier. In addition, the detection is quantized and the drift source is identified. SDDM quantifies the drift directly from the data, providing an explanation of the cause of the drift, so it is more useful in the real world than simply indicating whether the drift occurs.

$$T = (R - \alpha_R)/\beta_R \tag{20}$$

Hybrid forest method based on data distribution [69] is based on Hoeffding tree and can solve the problem of concept drift in data stream classification. This method uses a complete Hoeffding tree as the base classifier. This method benefits from the fast start-up ability and helps the system to predict faster than other methods at the beginning of data flow arrival. The OS-ELM model is quantified based on the extent to which the online sequential extreme learning machine (OS-ELMs) [70] is modified by updating the newly collected data. This method does not need any requirements of the amount of data collected, does not need the knowledge of the probability distribution of data streams, and is not based on hand-made statistical features, but proposes a process for automatically setting the dynamic threshold for concept drift detection. The Euclidean distance is shown in the Eq. (21). Among them, the output weight $\phi$, a measure of the difference between the baseline model $R'(a_1)$ trained data using time collected before $a_1$ and the data update model $R'(a_1 + \Delta a)$ collected between time $a_1$ and time $a_1 + \Delta a$, $d(\phi_{a_1}, \phi_{a_1+\Delta a})$ is the distance between the matrix $\phi_{a_{1\,m,n}}$ and $\phi_{a_1+\Delta a_{m,n}}$. This method updates the prediction model when the concept drift is detected to allow the overall accuracy of the energy prediction model to be improved and at the same time minimize the number of model updates. Density change recognition of nearest neighbor (NN-DVI) [56] is based on regional density estimation. It consists of three parts, namely, a space partition model based on k-nearest neighbor, which converts unmeasurable discrete data into a group of shared subspaces for density estimation. accumulate the density differences in these subspaces and quantify the distance function of the total difference and the confidence interval statistical significance test that can determine the concept drift.

$$D(R'(a_1), R'(a_1 + \Delta a)) = \sqrt{\sum_{m=1}^{H}\sum_{n=1}^{S}\left(\phi_{a_{1\,m,n}} - \phi_{a_1+\Delta a_{m,n}}\right)^2} \tag{21}$$

In addition, there is an online learning system: adaptive Classifier Integration system (ACE) [71], which is designed to deal with the changing environment and concept drift. The system consists of an online learner, a batch learner and a drift detection mechanism. The online learner learns from the new training sample in real time, while the batch learner forms a new hypothesis when it detects the concept drift. The system uses weighted majority voting to integrate the hypothetical output, uses a priori knowledge and responds quickly to abrupt changes. The system forms the final hypothesis by using a hypothetical $\{H_i\}_{i=0}^{I}$ weighted majority vote. The final assumption of the system is shown in Eq. (22). Among them $\alpha$ is the adjustment factor of the ensemble. The weight of each hypothesis is proportional to its applicability and increases exponentially with the increase of applicability. By using the applicability of each hypothesis in the weighted multiple voting, the system avoids the interference from the old classifier and makes use of the a priori knowledge of the recurrent context. Here, the selection of the hypothesis most suitable for the recent example is greatly improved by $\alpha$, while the smaller $\alpha$ increases the degree of the ensemble. Therefore, $\alpha$ should be chosen according to the complexity of the problem and the level of noise. The two variants MWDDM_H and MWDDM_M proposed by the multi-stage weighted drift detection method (MWDDM) [42] can detect abrupt and gradual concept drift faster than other comparison methods while keeping low false positive rate and false positive rate.

$$H_c(d_n) = \underset{x \in X}{\operatorname{argmax}} \sum_{i=0}^{I}\left(\frac{1}{1 - B_{i,n}}\right)^{a}[H_i(d_n) = x] \tag{22}$$

#### 4.1.2. Abrupt, gradual and incremental concept drift

For conceptual drift in data streams with abrupt drift, gradual drift and incremental drift, there are integration-based methods, data distribution-based methods, quaternion-based methods and weighting-based methods. Based on the integrated adaptive random forest (ARF) [72] method, the changing data streams are classified. Compared with previous attempts to copy random forests for data flow learning, ARF includes effective recovery methods and adaptive operators that can identify different types of concept drift without complex optimization for different data sets. When an instance in the data stream is available, each base tree is trained according to the RFTree Train function, and the warning or drift threshold and the corresponding warning or drift detection method are set. When the warning detection method finds a warning, a "background" tree is created for the tree, which is trained along the integration like the base tree and does not invoke the background tree without detecting the actual drift. If a drift is detected for the tree that generates the warning signal, the background tree will replace the corresponding base tree. The integration-based online diversification drift detection method (EN—ODDD) [73] is equipped with dynamic update integration to speed up adaptation to constantly changing distributions. This method requires the classifier to use different random input data subsets. The block-based classifier ensemble method includes active weighted aging enhancement method (Active Weighted Aging Ensemble, AWAE) [74], which is suitable for the classification of non-stationary data streams and is a weighted and pruning classifier in the set. The pruning process of classifier based on synthesis criterion is used, which considers the influence of given basis classifier on the diversity and accuracy of ensemble, and proposes an active learning method with finite label budget, which can be applied to any block-based classifier integration. AWAE has a built-in mechanism to ensure a limited ensemble size. If this value is exceeded, a classifier-based pool with the best evaluation function value is selected. Pruning is possible when the classifier $L$ is added to the integration, or after determining the weight used by the combination rules. When selecting post-pruning, use it accurately as shown in Eq. (23). Among them, $Q_1(\Pi)$ represents the accuracy of the classifier set, and $div(\Pi)$ represents the diversity of the classifier set, and the user-defined parameters used by the linear combination of accuracy and diversity satisfy. These two metrics are calculated based on the incoming data blocks. It adopts the square method of classifier integration, and uses the ever-changing integrated members to detect concept drift.

$$cri_q(\Pi) = \eta Q_1(\Pi) + (1-\eta) div(\Pi) \tag{23}$$

Error contribution weighting (ECW) and gradient descent weighting (GDW) [75] are based on the concept of continuous adaptive weighting. These methods use two prediction models, which are trained with the latest series and all series respectively. Finally, the weighted average of the predictions provided by the two models is considered to be the final prediction. In ECW, the percentage of double bias error $E_{p_i}$ and the percentage of double total error of bilinear pretest$E_{a_i}$ are defined as shown in Eqs. (24) and (25), respectively. Therefore, the final weighted prediction formula corresponding to the time point is: $\widehat{y}_i = E_{a_{i-1}} \cdot \widehat{y}_{partial_i} + E_{p_{i-1}} \cdot \widehat{y}_{all_i}$. And the final prediction formula of GDW is: $\widehat{y}_i = w_{p_i} \cdot \widehat{y}_{partial_i} + w_{a_i} \cdot \widehat{y}_{all_i}$. Where $w_{p_i} = w_{p_{i-1}} - g_{p_{i-1}}$ and $w_{a_i} = w_{a_{i-1}} - g_{a_{i-1}}$, $\eta$ is the adjusted rate coefficient. Finally, the weighted average of the predictions provided by the two models is considered to be the final prediction. The tag correlation drift detector (LD3) [13] method based on data distribution uses the prediction of dependencies between tags. It works with any online multi-tag classifier that does not have inherent drift detection attributes to help deal with changes in data distribution. The change of the tag dependency is evaluated by the prediction tag provided by the pairing model, and the concept drift will be detected if the dependency changes significantly. There is also a method based on quaternions to understand conceptual drift in data streams (QuadCDD) [76] not only detects and predicts the starting point of conceptual drift, but also uses quaternions to provide a more detailed analysis of conceptual drift, including drift start, drift end, drift severity, and drift type. The framework uses quaternions to make wise decisions and take appropriate actions to deal with various concept drifts to effectively maintain high and stable performance in data streams with conceptual drift.

$$E_{p_i} = \frac{\left(y_i - y_{partial_i}\right)^2}{\left(y_i - y_{partial_i}\right)^2 + \left(y_i - y_{all_i}\right)^2} \tag{24}$$

$$E_{p_i} = \frac{\left(y_i - y_{all_i}\right)^2}{\left(y_i - y_{partial_i}\right)^2 + \left(y_i - y_{all_i}\right)^2} \tag{25}$$

#### 4.1.3. Abrupt, gradual and recurrent concept drift

Most of the processing methods for drift data streams with abrupt, gradual and recurrent concepts are integration-based methods, such as Boosting-based integration, online integration-based and deep integration-based methods. Online lifting based on adaptive diversity (ADOB) [15] is an integration method based on online lifting method. The method adjusts the accuracy of each base classifier by controlling the diversity of allocation examples and uses an adaptive window as a drift detector, as shown in Eq. (26). Among them, $\mu_n = \frac{\theta_n}{1-\theta_n}$ and $\theta_n = \left(\frac{v_n^{sw}}{v_n^{sc}+v_n^{sw}}\right), m \in [1,n]$ is limited by the number of classifiers. When ADWIN returns a warning, a new set is established and trained together, a new ensemble is created after confirming the drift and the ensemble representing the last distribution is deleted. Knowledge maximization Integration (KME) [77] uses supervised and unsupervised knowledge to detect concept drift, identify recurrent concepts, and evaluate the weight of integration members. Instances of tags retained in past blocks can be reused to improve the recognition ability of candidate members. The final decision for the input observation is obtained from all the prediction results of the component classifier. Therefore, we can make the best use of the relevant information in the data stream. KME can respond to many types of concept drift by combining online and chunking-based integration mechanisms. BOLE [43] is improved on the basis of ADOB and OzaBoost. This method is effective for all types, but the integrated diversity of unlabeled observations needs to be improved. Flow deep forest (SDF) [78] square method, as a high-performance deep integration method, has a good detection effect for various drift types, and suitable for flow classification. At the same time, SDF is the first deep integration model used under the data flow setting, which uses the layer-by-layer processing of the original characteristics to achieve higher prediction accuracy.

$$h(\alpha) = \underset{\beta \in B}{\operatorname{argmax}} \sum n : h_n(\alpha) = \beta^{\log \frac{1}{\mu_n}} \tag{26}$$

#### 4.1.4. Real and virtual concept drift

For data streams with virtual drift and various real types of concept drift, there are methods based on Gaussian mixture model and neural network.

The Gaussian mixture model (GMM-VRD) [79] method deals with virtual drift by updating and creating the Gaussian model, and real drift by resetting the system. Use the higher values of the AIC criteria to select the best result model and add it to the final GMM model. The AIC criteria are defined as follows: $AIC = 2I - 2J$. In a GMM with only one Gaussian, $I$ represents the number of model parameters, where parameters mean covariance and weight. Therefore, for each existing Gaussian in GMM, the value of $I$ is multiplied by 3. Parameter $J$ represents the maximum likelihood function of GMM, as shown in Eq. (27). Where $N$ represents the number of observations used for training, $M$ represents the Gaussian number in the GMM model, which $x_a$ is a multivariable observation, which $\varphi_b$ represents the weight of the number of observations that make up the mixed component $b$, $\sum_{b=1}^{M} \varphi_b = 1$ and $0 \leq \varphi_b \leq 1$, where $P(x_a|C_b)$ represents the conditional probability that the observation $x_a$ belongs to the mixed component $C_b$. In terms of accuracy over time, the method shows lower conceptual drift degradation, which shows that the proposed method is effective. Another method to solve these two drift problems is called on-line Gaussian mixture model and noise filter to deal with virtual and real concept drift (OGMMF-VRD) [60]. Noise filters can avoid adapting to observations that may cause problems in the system, and online learning can

maintain the existing Gaussian distribution by modeling the new distribution. Using correlation threshold, we can quickly create Gaussian function to deal with virtual drift correctly, and use pool to make it robust to false positives in drift detection caused by noise examples. In order to solve the drift problem in the data stream with virtual drift and real drift, the method based on neural network has integrated self-construing Ensemble Self-Structuring Neural Network (ESSNN) [58]. This method is more inclined to virtual concept drift, in which the important characteristics used to predict class variables may change over time.

$$J = \sum_{a=1}^{N} \log \sum_{b=1}^{M} P(x_a|C_b) \cdot \varphi_b \tag{27}$$

### 4.2. Group type

Dealing with group concept drift usually requires different treatment strategies for different groups. For data streams with group concept drift, there are methods based on data distribution, methods based on time warps and adaptive methods based on distribution.

#### 4.2.1. Data distribution-based method

At present, there are only several methods to detect group drift, such as the group drift detection method (GDDM) [3]. It inherits the idea of the drift detection method based on error rate for a single data stream, that is, the input variable of GDDM is the error rate rather than the data itself, in order to ignore the difference in the number and scale of features. The difference is that the input variables in GDDM are multivariable because the error rates of all data streams are considered at the same time. In addition, in order to ignore the potential distribution of data streams and the correlation between data streams, the method proposes a new test statistic, which assumes that $\widehat{F}_{a+b}(\theta)$ represents the tested distribution function $\{L_{t+1}, \ldots, L_{t+u}, \ldots, L_{t+a}, L_{t+a+1}, \ldots, L_{t+a+v}, L_{t+a+b}\}$ for merged samples $\theta \in R^N$, as shown in Eq. (28). Where $L_P$ is the indicator of the event $P$ and $\theta$ is the user-defined threshold. For stream data, an online learning method is designed to obtain the test statistics, so as to determine the concept drift caused by hypothesis testing.

$$F_{a+b}(\theta) = \frac{\sum_{u=1}^{a} L_{W_{u+1} \le \theta} + \sum_{v=1}^{b} L_{W_{u+b+v} \le \theta}}{a+b} \tag{28}$$

#### 4.2.2. Time-warping-based method

The concept drift learning method based on time warp (TW-CDM) [80] is used to deal with multi-data streams. This method identifies the drift of multi-data streams in real time, and trains the model on the data block and tests it. The drift is detected by measuring the similarity of the sequence data between two data streams at a certain time step. The matching path obtained by calculating the distance between the two streams as shown in the Eq. (29) shows that it is similar, that is, $\rho^{A_1}_{m+1,x} \leftarrow \rho^{A_2}_{m+1,y}$ or $\rho^{A_1}_{m+1,x} \leftarrow \left\{\rho^{A_2}_{m+1,y}, \ldots, \rho^{A_2}_{m+1,y+n}\right\}$. The reduced precision of $\rho^{A_1}_{m+1} = \left\{\rho^{A_1}_{m+1,x}\right\}^{k}_{x=1}$ and $\rho^{A_2}_{m+1} = \left\{\rho^{A_2}_{m+1,y}\right\}^{k}_{y=1}$ is calculated to identify the drift in $A_1$ and the possible drift area of $A_2$. The training model is updated by cross-selecting the data between the two data streams for reinforcement learning.

$$dis\left(\rho^{A_1}_{m+1}, \rho^{A_2}_{m+1}\right) = \left|\rho^{A_1}_{m+1} - \rho^{A_2}_{m+1}\right| \tag{29}$$

#### 4.2.3. Distribution based method

Using adaptive methods to solve group-specific distributed concept drift, FairFedDrift [81] adopts multi-model strategy, local group-specific drift detection mechanism and continuous clustering of the model over time. This method uses global loss monitoring to handle group-specific concept drift. There are different global models in the global model set, the distributed cooperative training model, and there is a global model that is related to each client. If there is no global model in the set, the loss difference of the two consecutive timestamps in the set is smaller than the threshold. As shown in Eq. (30), it represents drift generation. Where $\forall \lambda_n \in Global : \forall l : w^t_l = m : \forall a \in A : \xi(\lambda_n)^t_{a,l} \le \xi(\lambda_n)^{t-1}_{a,l} + \sigma_a$, $t \in T$ represents time step, $l \in L$ represents the clients within the federation, $s \in S$ represents the sensitive group, and $\lambda_n \in Global$ represents the specific global model, $l \in L : w^t_l = m$ is a collection of clients identified with the global model $\lambda_n$ at time step $t$, $f(\lambda_n)^t$ is the weighted sum of the client's local targets $G_l(\lambda)$ identified by the global model $\lambda_n$ at time step $t$, and $\xi(\lambda_n)^t_{a,l}$ targets is the loss of group $s$ and client $l$ on the model $\lambda_n$ steps at time step $t$. $\sigma_a$ represents the predefined threshold for the acceptable loss difference of group $s$ between two continuous timestamps on the global model.

$$\min_{\lambda_n \in Global} f(\lambda_n)^t = \sum_{l \in L: w^t_l = n} \frac{|D^t_l|}{|D^t|} G_l(\lambda), w^t_l = \arg\min_{a} \sum_{a \in A} \xi(\lambda_n)^t_{a,l} \tag{30}$$

### 4.3. Analysis and summary

This section summarizes and analyzes the methods used in Chapter Four, and explains them from the perspectives of applicable types, core ideas, experimental datasets, comparison methods, and advantages and disadvantages, providing a more vivid summary of

the methods in Chapter Four. In addition, the classification accuracy results of six methods, namely ADWIN, DDM, EDDM, HDDM, FHDDM, OCDD and SDF, on nine datasets are listed, visually demonstrating the applicable scenarios of different methods.

The methods listed in this chapter to deal with abrupt and gradual drift data streams are based on integrated ACE method, statistical WSTD, Plover, SDDM method, window-based FHDDM, OCDD, CDT_MSW, MWDDM method, OS-ELMs, Hybrid Forest and NN-DVI method based on data distribution. The data flow methods with abrupt change, gradual change and incremental drift include integrated ARF, AWAE and En-ODDD methods, weighted ECW and GDW methods, LD3 method based on data distribution and QuadCDD method based on quaternion. Data flow methods with abrupt change, gradual change and repeated drift are based on integrated KME,

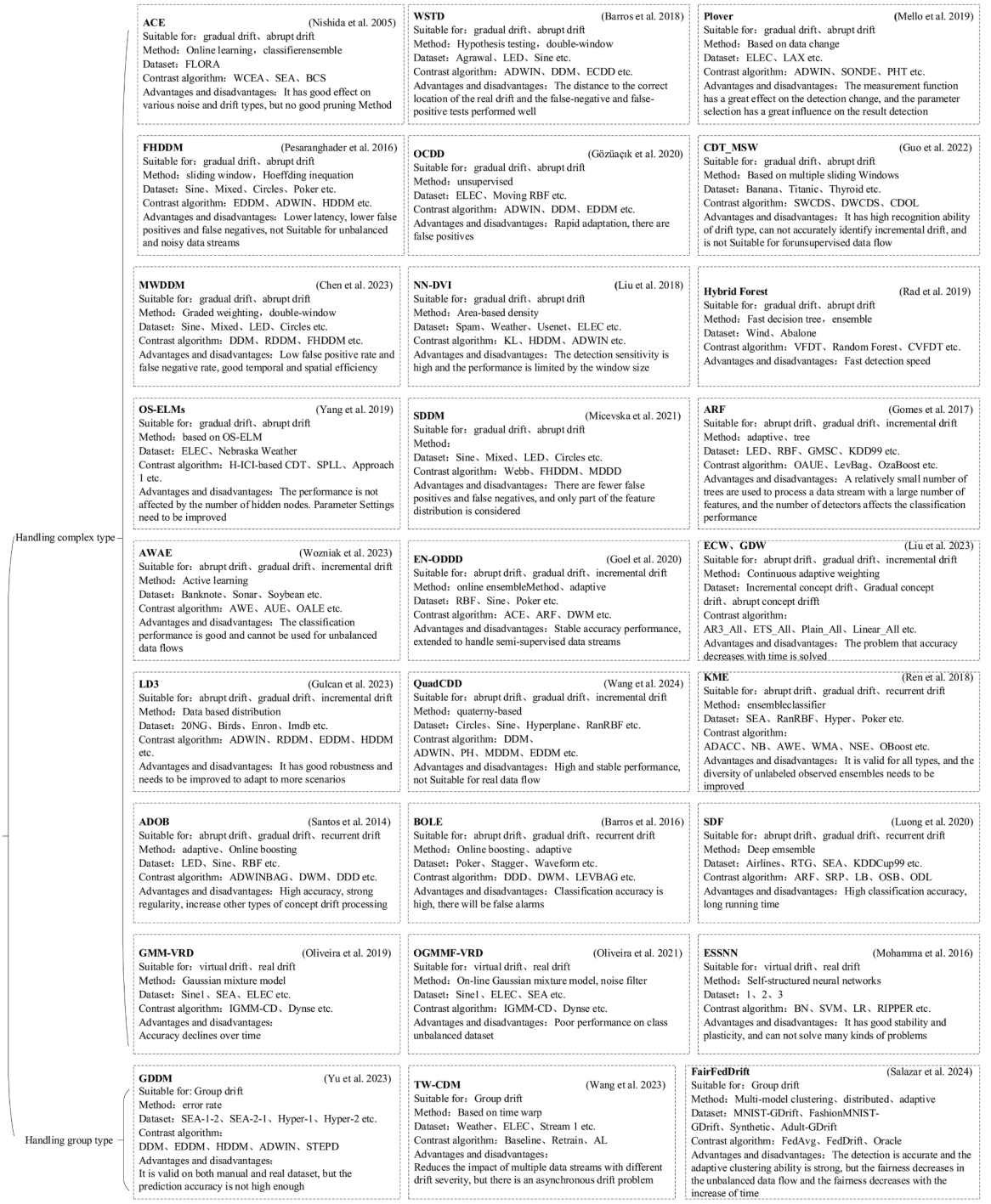



**Fig. 7.** Summary of methods for dealing with multiple type and group type.

ADOB, SDF and BOLE methods. GMM-VRD and OGMMF-VRD methods based on Gaussian mixed model and ESSNN method based on neural network are used to deal with data streams with virtual and real drift. The methods to deal with group drift are statistics-based GDDM, time warp-based TW-CDM and distributed FairFedDrift. The methods analyzed in this chapter for dealing with compound types and group types are summarized as shown in Fig. 7.

According to literature [53,54] and [66], nine data sets of ELEC, Covtype, Poker Hand, Outdoor, Airlines, Spam, Phissing, Hyperplane and RBF are compared, in which Hyperplane and RBF are artificial data sets, and ELEC, Covtype, Outdoor, Spam, Phissing, Airlines and Poker Hand are real data sets. The comparison methods include the classical ADWIN, DDM, EDDM and RDDM methods, and the experimental results are shown in Fig. 8. In the ELEC data set, the average accuracy of OCDD is higher than that of other methods with an average accuracy of 86.22 %, especially that of EDDM. In the RBF data set, FHDDM and OCDD are significantly higher than ADWIN, DDM and EDDM, especially compared with EDDM square method, the average accuracy of EDDM method is 1.63times and 1.62times, respectively. The average accuracy of OCDD on Covtype and Outdoor data sets is 88.36 % and 62.11 %, respectively, higher than that of other comparison methods, and the average accuracy of 90.56 % on Phissing data sets is slightly higher than that of other comparison methods. No matter what method is used in Phissing and Hyperplane datasets, it has good accuracy. SDF has an average accuracy of 95.66 % on Covtype data sets, which is 1.19 times higher than that of ADWIN method. SDF and GDDM have an average accuracy of 68.57 % and 70.71 % on Airlines data sets, respectively, and GDDM is the only square method with an average accuracy of more than 70 %. GDDM has an average accuracy of 91.23 % on Hyperplane data sets, which is 10.96 % higher than that of EDDM method. In RBF data sets, the average accuracy of FHDDM and GDDM is 98.08 % and 97.02 % respectively, which is 38.02 % and 36.96 % higher than that of EDDM method. The EDDM method does not achieve an average accuracy of 90 % on all data sets, and only 82.76 % on Covtype data sets.

## 5. Results

This chapter mainly conducts a systematic analysis of the methods listed in Chapters Three and Four, including the applicable drift types and data flow characteristics. On this basis, discuss its application in real-world scenarios.

Through the analysis in the previous two chapters, it can be found that the existing methods have been able to solve most problems. For instance, methods such as ADWIN [11], MDDT [40], and MDDM [41] have advantages in dealing with mutation drift. Methods such as RDDM [46], FW-DA [48], EnHAT [49] and VDD [50] are more suitable for solving the problem of gradual drift. Methods such as CPF [51], ECPF [52] and RDWM [53] can handle the problem of repeated drift in data streams very well. DiwE [16], LDD-DSDA [55] and NN-DVI [56] can effectively target local drift. Methods such as DEnBoost [38], IGMM-CD [59], OGMMF-VRD [60], Dynse [61] and VDDM [62] can handle virtual drift relatively well. It is worth noting that CDT-MSW [12], MWDDM [42], ADOB [15] and BOLE [44] can not only solve a single type of drift problem, but also handle data streams containing multiple types of drift problems.

The above-mentioned methods only address the concept drift problem in a single data stream. However, to solve the group concept drift problem in multiple data streams, methods such as GDDM [3], TW-CDM [80], and FairFedDrift [81] have been proposed. GDDM uses the most classic error-rate-based approach to detect group drift, while FairFedDrift performs group drift detection through data distribution. TW-CDM uses the data similarity within the time step to determine whether group drift has occurred. In addition, more and more new methods are also being studied.

Overall, there are already quite a few concept drift detection methods to choose from for solving different concept drift problems [82]. For example, LD3 [13], EDDM [9], OGMMF-VRD [60] and ACE [71] can solve the problem of concept drift in noisy data streams. However, due to the reduced detection sensitivity, the detection delay is relatively high. MWDDM, WSTD [67], SDDM [14] and

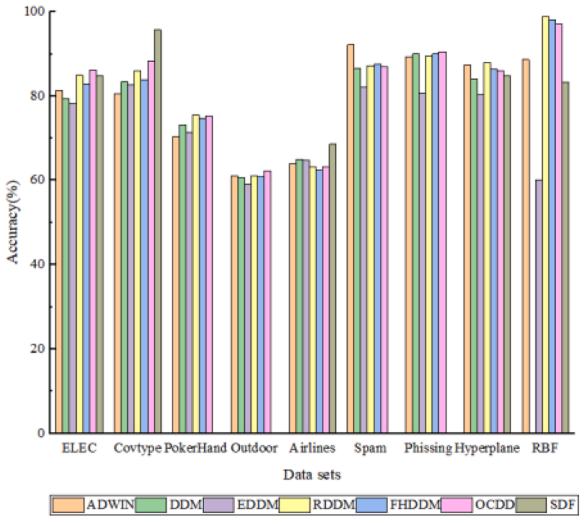



**Fig. 8.** Average accuracy of seven methods compared with classical methods on different data sets.

FHDDM [65] have relatively low false alarm rates and poor robustness when detecting drift. Combining the characteristics of the methods, many methods can be applied to practical scenarios. For example, CDDA-MD [5] has been used for malicious traffic detection, while M2FD [6] can conduct joint detection of malicious software. In addition, a dynamic integration method [31] has been developed and applied in the field of antibiotic resistance in hospitals. Also, An effective framework for estimating the generalization error of regression function [63] is applied to phishing websites.

## 6. Future work

Although some progress has been made in the processing methods for concept drift in data streams, concept drift is a complex issue and still faces some challenges. Future research directions include:

1) Research on window-based group concept drift detection method

Existing work has been done to provide accurate predictions for online data streams by updating and adjusting models or by retraining the latest local data through sliding Windows, but this approach only works for a single data stream. In the real world, data flows are basically not independent, but multiple related data flows exist simultaneously. The problem of concept drift across multi-data streams is further complicated by the need to perform data analysis across multi-data streams simultaneously when making predictions. A large number of data streams are constantly updated, and setting a separate concept drift processing module on each data stream will not only multiply the amount of computation, but also greatly extend the time required for the prediction task [83]. Concept drift in multi-data streams is a problem that has been raised in recent years, and there are few researches on the concept drift of group type. However, compared with the detection of concept drift in a single data stream, there are three challenges in solving the group concept drift problem: the training data is more complex, the underlying distribution of the data stream is more complex, and the correlation of the data stream is more complex. The author intends to combine multi-data streams into one data stream and use a fixed-size window to detect concept drift. By calculating the average distance and standard deviation between the two errors, the distance between the two errors is obtained. When the proportion of the maximum distance distribution between one error and the other is less than a certain threshold, it means that the drift has occurred.

2) Research on concept drift in multi-class unbalanced data streams

Non-stationary data flows lead to changes in data characteristics, leading to concept drift and class imbalance [84]. Most classifiers are only applicable when the classes are roughly balanced, otherwise the classifier will be biased against the majority of classes, thus reducing the generalization ability of the classifier. At present, the main methods to alleviate the class imbalance problem are to reduce the majority of classes or increase the minority of classes by training the data set or reordering, or to complete the model adaptation by modifying the classifier used. A class in multiclass unbalanced data can be a majority of some classes and a minority of others. Multi-class unbalance problems are more common in the real world, but most of the existing conceptual drift processing methods focus on binary unbalance problems. The relationship between classes in multi-class unbalanced data is more complex, and its decomposition into binary imbalance problem will lead to information loss and oversimplification, so it is necessary to design a concept drift processing method suitable for multi-class unbalanced data [85]. The next step is to analyze the instance-level difficulties of each class, so as to understand the causes of learning difficulties, and design an integrated model that can deal with multi-class imbalance problems.

3) The study of group concept drift in noisy data streams

When a new instance arrives, the current model may become outdated due to the possible gradual changes between the two distributions. At this point, it is rather difficult to find the exact distribution change points and it is also necessary to distinguish the noise that may affect the stream. Therefore, drift detection methods must be robust and sensitive to real changes in the data set, striking a balance between stability and plasticity [86]. Moreover, local drift will sometimes be considered noise and will lead to model instability, so the model must be able to effectively distinguish local variation from noise and handle the scarcity of instances representing drift in order to update the learner more efficiently. Window methods assume that drift affects the entire instance space uniformly, so they cannot detect local drift that changes only in certain regions of the instance space. At the same time, the window method cannot handle the scarcity of instances representing drift and treats them as noise until the time for detecting local conceptual drift can be arbitrarily long [87]. However, for noisy data streams, the group concept drift with correlation is more complex. The author intends to improve the existing window method or local drift detection method and increase the robustness of the method to obtain a more efficient and accurate group concept drift processing method.

## 7. Conclusion

Concept drift, as an important problem in supervised learning tasks, has attracted extensive research attention. This paper introduces the definition, causes and types of concept drift. In response to changes in data distribution due to environmental changes, data acquisition biases or changes in the concept itself, researchers have proposed a number of methods to deal with concept drift, each of which has advantages and disadvantages and is suitable for different drift types and application scenarios. With the introduction of multi-data streams in recent years, more and more attention has been paid to the study of group concept drift. This paper classifies the methods to deal with concept drift from three perspectives, which are for single type, for compound type and for group type. The processing methods for a single type are divided into real type concept drift processing methods and virtual type concept drift processing methods. The real type concept drift processing methods are mainly abrupt type, gradual type, recurrent type and other types. The treatment methods for compound types are divided into abrupt and gradual type, abrupt, gradual type and incremental type, abrupt, gradual type and repeated type and other compound types. The processing methods of group type concept drift are analyzed,

and the applicable types, main techniques, experimental data sets, comparison methods, advantages and disadvantages of the analyzed algorithms are summarized. Based on the existing processing methods and research status, the future research direction is mainly to study the window-based detection method for the concept drift problem of group type, and solve the concept drift problem in multi-class unbalanced data streams and noisy data streams.

## CRediT authorship contribution statement

**Shurong Yang:** Data curation. **Meng Han:** Formal analysis, Conceptualization. **Shineng Zhu:** Supervision. **Wenyan Yang:** Formal analysis. **Zhenlong Dai:** Supervision. **Juan Li:** Formal analysis. **Jian Ding:** Supervision.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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